

E-Commerce Customer Churn Analysis

Data Cleaning

Handling Missing Values and Outliers:

➤ Impute mean for the following columns, and round off to the nearest:

WarehouseToHome column;

```
5666 • SELECT ROUND(AVG(WarehouseToHome))FROM customer_churn;
```

```
5667 • UPDATE customer_churn SET WarehouseToHome =16 WHERE WarehouseToHome :
```

```
5668 • SET SQL_SAFE_UPDATES = 0;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

ROUND(AVG(WarehouseToHome))
16

```
5666 • SELECT ROUND(AVG(WarehouseToHome))FROM customer_churn;
```

```
5667 • UPDATE customer_churn SET WarehouseToHome =16 WHERE WarehouseToHome IS NULL;
```

```
5668 • SET SQL_SAFE_UPDATES = 0;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows: |

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus	NumberOf...
50044	0	11	Mobile Phone	3	6	Debit Card	Female	2	4	Laptop & Accessory	2	Single	5
50045	0	5	Phone	1	12	Debit Card	Male	NULL	3	Mobile	3	Single	4
50046	0	0	Phone	1	23	COD	Male	NULL	3	Mobile	4	Divorced	1
50047	1	0	Computer	3	16	COD	Female	3	5	Mobile	5	Single	1
50048	0	NULL	Computer	1	9	CC	Male	1	3	Mobile	4	Single	5

HourSpendOnApp column;

```
5670 • SELECT ROUND(AVG(HourSpendOnApp))FROM customer_churn;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

ROUND(AVG(HourSpendOnApp))
3

5661 • `SELECT * FROM customer_churn;`
5662

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus
50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Accessory	2	Single
50002	1	NULL	Phone	1	8	UPI	Male	3	4	Mobile	3	Single
50003	1	NULL	Phone	1	30	Debit Card	Male	2	4	Mobile	3	Single
50004	1	0	Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory	5	Single
50005	1	0	Phone	1	12	CC	Male	NULL	3	Mobile	5	Single

5670 • `SELECT ROUND(AVG(HourSpendOnApp)) FROM customer_churn;`
5671 • `UPDATE customer_churn SET HourSpendOnApp = 3 WHERE HourSpendOnApp IS NULL;`

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus
50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Accessory	2	Single
50002	1	NULL	Phone	1	8	UPI	Male	3	4	Mobile	3	Single
50003	1	NULL	Phone	1	30	Debit Card	Male	2	4	Mobile	3	Single
50004	1	0	Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory	5	Single
50005	1	0	Phone	1	12	CC	Male	3	3	Mobile	5	Single

OrderAmountHikeFromLastYear column;

5673 • `SELECT ROUND(AVG(OrderAmountHikeFromLastYear)) FROM customer_churn;`
5674
5675
5676
5677

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
ROUND(AVG(OrderAmountHikeFromLastYear))			
16			

When the NULL value is present

HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress	Complain	OrderAmountHikeFromLastYear	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount
3	4	Mobile	3	Divorced	2	1	14	0	1	2	127
2	5	Mobile	3	Single	2	0	12	1	1	1	123
2	3	Others	3	Divorced	2	0	NULL	9	15	8	295
3	4	Fashion	3	Single	10	1	13	0	1	0	154
2	3	Mobile	3	Single	2	1	13	2	2	2	134

After the NULL value is replaced by the rounded average value;

HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress	Complain	OrderAmountHikeFromLastYear	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount
5	2	Mobile	2	Single	5	1	13	5	5	3	134
3	4	Mobile	3	Divorced	10	1	13	0	1	0	124
5	2	Others	3	Divorced	5	0	16	8	12	8	582
5	2	Mobile	3	Single	5	0	15	1	1	1	153
2	4	Mobile	2	Divorced	5	1	14	0	1	5	153

5678 • `UPDATE customer_churn SET OrderAmountHikeFromLastYear = 16 WHERE OrderAmountHikeFromLastYear IS NULL;`
5679 • `SELECT ROUND(AVG(OrderAmountHikeFromLastYear)) FROM customer_churn;`

DaySinceLastOrder column;

```
5677 • SELECT ROUND(AVG(DaySinceLastOrder))FROM customer_churn;
```

```
5678 • UPDATE customer_churn SET DaySinceLastOrder =5 WHERE DaySinceLastOrder IS NULL;
```

```
5679
```

```
5680
```

```
5681
```

HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress	Complain	OrderAmountHikeFromlastYear	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount
2	3	Laptop & Accessory	3	Single	2	0	14	1	1	7	200
2	4	Fashion	4	Divorced	6	0	16	1	4	5	225
2	3	Mobile	5	Single	6	1	11	1	1	4	127
3	3	Fashion	2	Single	4	1	18	0	0	0	236
2	4	Laptop & Accessory	5	Single	6	0	15	0	1	1	167

➤ Impute mode for the following columns: Tenure, CouponUsed, OrderCount.

Tenure column;

In Tenure column the mode value is 1

```
5680 /*Impute mode for the following columns: Tenure*/
```

```
5681 • SELECT Tenure, COUNT(*) AS tenure_mode FROM customer_churn WHERE Tenure IS NOT NULL GROUP BY Tenure ORDER BY tenure_mode DESC LIMIT 1;
```

```
5682 • UPDATE customer_churn SET Tenure=1 WHERE Tenure IS NULL;
```

```
5683
```

```
5684
```

Tenure	tenure_mode
1	954

The NULL value in the Tenure column is filled with 1

```
5682 • UPDATE customer_churn SET Tenure=1 WHERE Tenure IS NULL;
```

```
5683
```

```
5684
```

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus	Number
50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Accessory	2	Single	9
50002	1	1	Phone	1	8	UPI	Male	3	4	Mobile	3	Single	7
50003	1	1	Phone	1	30	Debit Card	Male	2	4	Mobile	3	Single	6
50004	1	0	Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory	5	Single	8
50005	1	0	Phone	1	12	CC	Male	3	3	Mobile	5	Single	3

CouponUsed column;

Before handling null values in the CuponUsed column

HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress	Complain	OrderAmountHikeFromlastYear	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount
2	2	Laptop & Accessory	5	Married	2	0	11	1	1	3	145
3	3	Mobile	4	Divorced	2	0	13	0	1	6	128
2	3	Others	3	Divorced	2	0	14	0	9	9	290
3	2	Others	4	Single	9	0	11	2	2	15	287
2	4	Laptop & Accessory	3	Divorced	2	0	17	0	1	9	195

The mode value of CuponUsed column is 1, so the null value is replaced with 1.

```
5684 /*Cuponused*/
5685 • SELECT CuponUsed, COUNT(*) AS cupon_mode FROM customer_churn WHERE CuponUsed IS NOT NULL GROUP BY CuponUsed ORDER BY cupon_mode DESC LIMIT 1;
5686 • UPDATE customer_churn SET CuponUsed=1 WHERE CuponUsed IS NULL;
```

HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress	Complain	OrderAmountHikeFromLastYear	CuponUsed	OrderCount	DaySinceLastOrder	CashbackAmount
2	3	Others	3	Divorced	2	0	14	1	9	9	290
3	2	Others	4	Single	9	0	11	1	2	15	287
2	4	Laptop & Accessory	3	Divorced	2	0	17	0	1	9	195
3	3	Grocery	4	Single	4	0	18	0	1	0	237
3	4	Others	3	Divorced	1	0	16	1	11	9	300

OrderCount column;
Before handling NULL value

HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress	Complain	OrderAmountHikeFromLastYear	CuponUsed	OrderCount	DaySinceLastOrder	CashbackAmount
2	3	Others	3	Divorced	2	0	14	1	9	9	290
3	2	Others	4	Single	9	0	11	1	2	15	287
2	4	Laptop & Accessory	3	Divorced	2	0	17	0	1	9	195
3	3	Grocery	4	Single	4	0	18	0	NULL	0	237
3	4	Others	3	Divorced	1	0	16	1	11	9	300

In OrderCount column we get mode value is 2;

```
5688 /*Ordercount*/
5689 • SELECT OrderCount, COUNT(*) AS order_mode FROM customer_churn WHERE OrderCount IS NOT NULL GROUP BY OrderCount ORDER BY order_mode DESC LIMIT 1;
```

OrderCount	order_mode
2	2025

After handling NULL value by mode value, here mode value is 2

```
5690 • UPDATE customer_churn SET OrderCount=2 WHERE OrderCount IS NULL;
```

HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress	Complain	OrderAmountHikeFromLastYear	CuponUsed	OrderCount	DaySinceLastOrder	CashbackAmount
2	4	Laptop & Accessory	3	Divorced	2	0	17	0	1	9	195
3	3	Grocery	4	Single	4	0	18	0	2	0	237
3	4	Others	3	Divorced	1	0	16	1	11	9	300
2	3	Mobile	4	Single	2	0	16	1	1	3	123
2	4	Laptop & Accessory	3	Divorced	4	1	13	1	1	1	159

➤ Handle outliers in the 'WarehouseToHome' column by deleting rows where the values are greater than 100.

```
5689 /* Handle outliers in the 'WarehouseToHome' column by deleting rows where the values are greater than 100. */
5690 • DELETE FROM customer_churn WHERE WarehouseToHome>100;
```

#	Time	Action	Message
92	08:07:55	UPDATE customer_churn SET DaySinceLastOrder =5 WHERE DaySinceLastOrder IS NULL	307 row(s) affected Rows matched: 307 Changed: 307 Warnings: 0
93	08:07:57	SELECT Tenure, COUNT(*) AS tenure_mode FROM customer_churn WHERE Tenure IS NOT NULL GROUP BY...	1 row(s) returned
94	08:08:00	UPDATE customer_churn SET Tenure=1 WHERE Tenure IS NULL	264 row(s) affected Rows matched: 264 Changed: 264 Warnings: 0
95	08:08:03	SELECT CuponUsed, COUNT(*) AS cupon_mode FROM customer_churn WHERE CuponUsed IS NOT NULL...	1 row(s) returned
96	08:08:08	UPDATE customer_churn SET CuponUsed=1 WHERE CuponUsed IS NULL	256 row(s) affected Rows matched: 256 Changed: 256 Warnings: 0
97	08:08:14	SELECT OrderCount, COUNT(*) AS order_mode FROM customer_churn WHERE OrderCount IS NOT NULL GR...	1 row(s) returned
98	08:08:17	UPDATE customer_churn SET OrderCount=2 WHERE OrderCount IS NULL	258 row(s) affected Rows matched: 258 Changed: 258 Warnings: 0
99	08:08:30	SELECT *FROM customer_churn LIMIT 0, 1000	1000 row(s) returned
100	08:10:33	DELETE FROM customer_churn WHERE WarehouseToHome>100	2 row(s) affected

Dealing with Inconsistencies:

➤ Replace occurrences of “Phone” in the 'PreferredLoginDevice' column and “Mobile” in the 'PreferredOrderCat' column with “Mobile Phone” to ensure uniformity.

'PreferredLoginDevice' column

*Before handling the occurrences of “Phone” in the 'PreferredLoginDevice' column

	CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat
▶	50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Accessory
	50002	1	1	Phone	1	8	UPI	Male	3	4	Mobile
	50003	1	1	Phone	1	30	Debit Card	Male	2	4	Mobile
	50004	1	0	Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory

*After replace with “Mobile Phone”

```
5693 • UPDATE customer_churn SET PreferredLoginDevice='Mobile Phone' WHERE PreferredLoginDevice='Phone';
5694 • UPDATE customer_churn SET PreferredOrderCat='Mobile Phone' WHERE PreferredOrderCat='Mobile';
```

	CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat
▶	50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Accessory
	50002	1	1	Mobile Phone	1	8	UPI	Male	3	4	Mobile
	50003	1	1	Mobile Phone	1	30	Debit Card	Male	2	4	Mobile
	50004	1	0	Mobile Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory

'PreferredOrderCat' column

*Before handling the occurrences of “Mobile” in the 'PreferredOrderCat' column

	CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat
▶	50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Accessory
	50002	1	1	Mobile Phone	1	8	UPI	Male	3	4	Mobile
	50003	1	1	Mobile Phone	1	30	Debit Card	Male	2	4	Mobile
	50004	1	0	Mobile Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory

*After replace with “Mobile Phone”

```
5694 • UPDATE customer_churn SET PreferredOrderCat='Mobile Phone' WHERE PreferredOrderCat='Mobile';
```

	CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat
▶	50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Accessory
	50002	1	1	Mobile Phone	1	8	UPI	Male	3	4	Mobile Phone
	50003	1	1	Mobile Phone	1	30	Debit Card	Male	2	4	Mobile Phone
	50004	1	0	Mobile Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory

➤ Standardize payment mode values: Replace "COD" with "Cash on Delivery" and "CC" with "Credit Card" in the PreferredPaymentMode column.

*Before replacing "COD" with "Cash on Delivery";

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat
50010	1	1	Mobile Phone	1	31	Debit Card	Male	2	5	Mobile Phone
50011	1	4	Mobile Phone	1	18	Cash on Delivery	Female	2	3	Others
50012	1	11	Mobile Phone	1	6	Debit Card	Male	3	4	Fashion
50013	1	0	Mobile Phone	1	11	COD	Male	2	3	Mobile Phone

*After replacing with Cash on Delivery;

```
5697 • UPDATE customer_churn SET PreferredPaymentMode='Cash on Delivery' WHERE PreferredPaymentMode='COD';
```

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat
50010	1	1	Mobile Phone	1	31	Debit Card	Male	2	5	Mobile Phone
50011	1	4	Mobile Phone	1	18	Cash on Delivery	Female	2	3	Others
50012	1	11	Mobile Phone	1	6	Debit Card	Male	3	4	Fashion
50013	1	0	Mobile Phone	1	11	Cash on Delivery	Male	2	3	Mobile Phone

*Before replacing "CC" with "Credit Card";

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat
50002	1	1	Mobile Phone	1	8	UPI	Male	3	4	Mobile Phone
50003	1	1	Mobile Phone	1	30	Debit Card	Male	2	4	Mobile Phone
50004	1	0	Mobile Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory
50005	1	0	Mobile Phone	1	12	CC	Male	3	3	Mobile Phone

*After replacing with Credit Card;

```
5696 /*Standardize payment mode values: Replace "COD" with "Cash on Delivery" and "CC" with "Credit Card" in the PreferredPaymentMode column*
5697 • UPDATE customer_churn SET PreferredPaymentMode='Cash on Delivery' WHERE PreferredPaymentMode='COD';
5698 • UPDATE customer_churn SET PreferredPaymentMode='Credit Card' WHERE PreferredPaymentMode='CC';
```

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat
50002	1	1	Mobile Phone	1	8	UPI	Male	3	4	Mobile Phone
50003	1	1	Mobile Phone	1	30	Debit Card	Male	2	4	Mobile Phone
50004	1	0	Mobile Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory
50005	1	0	Mobile Phone	1	12	Credit Card	Male	3	3	Mobile Phone

Data Transformation:

Column Renaming:

➤ Rename the column "PreferredOrderCat" to "PreferredOrderCat"

```
5700 /*Rename the column "PreferredOrderCat" to "PreferredOrderCat"*/
5701 • ALTER TABLE customer_churn RENAME COLUMN PreferredOrderCat TO PreferredOrderCat;
```

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore
50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Accessory	2
50002	1	1	Mobile Phone	1	8	UPI	Male	3	4	Mobile Phone	3
50003	1	1	Mobile Phone	1	30	Debit Card	Male	2	4	Mobile Phone	3
50004	1	0	Mobile Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory	5

➤ Rename the column "HourSpendOnApp" to "HoursSpentOnApp"

```
5703 /*Rename the column "HourSpendOnApp" to "HoursSpentOnApp"*/
5704 • ALTER TABLE customer_churn RENAME COLUMN HourSpendOnApp TO HoursSpentOnApp;
5705
5706 • SELECT*FROM customer_churn;
```

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HoursSpentOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore
50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Accessory	2
50002	1	1	Mobile Phone	1	8	UPI	Male	3	4	Mobile Phone	3
50003	1	1	Mobile Phone	1	30	Debit Card	Male	2	4	Mobile Phone	3
50004	1	0	Mobile Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory	5

Creating New Columns:

➤ Create a new column named 'ComplaintReceived' with values "Yes" if the corresponding value in the 'Complain' is 1, and "No" otherwise.

*'ComplaintReceived' new column created;

NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress	Complain	OrderAmountLikeFromlastYear	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount	ComplaintReceived
3	Laptop & Accessory	2	Single	9	1	11	1	1	5	160	Yes
4	Mobile Phone	3	Single	7	1	15	0	1	0	121	Yes
4	Mobile Phone	3	Single	6	1	14	0	1	3	120	Yes
4	Laptop & Accessory	5	Single	8	0	23	0	1	3	134	No

* When the Complain=1 ,

```
5708 /*Create a new column named 'ComplaintReceived' with values "Yes" if the corresponding value in the 'Complain' is 1, and "No" otherwise*/
5709 • ALTER TABLE customer_churn ADD COLUMN ComplaintReceived VARCHAR(100);
5710 • UPDATE customer_churn SET ComplaintReceived = 'Yes' WHERE Complain = 1;
5711
```

NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress	Complain	OrderAmountLikeFromlastYear	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount	ComplaintReceived
3	Laptop & Accessory	2	Single	9	1	11	1	1	5	160	Yes
4	Mobile Phone	3	Single	7	1	15	0	1	0	121	Yes
4	Mobile Phone	3	Single	6	1	14	0	1	3	120	Yes
4	Laptop & Accessory	5	Single	8	0	23	0	1	3	134	No

*When the Complain=0'

```
5708 /*Create a new column named 'ComplaintReceived' with values "Yes" if the corresponding value in the 'Complain' is 1, and "No" otherwise*/
5709 • ALTER TABLE customer_churn ADD COLUMN ComplaintReceived VARCHAR(100);
5710 • UPDATE customer_churn SET ComplaintReceived = 'Yes' WHERE Complain = 1;
5711 • UPDATE customer_churn SET ComplaintReceived = 'No' WHERE Complain <> 1;
```

NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress	Complain	OrderAmountLikeFromlastYear	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount	ComplaintReceived
3	Laptop & Accessory	2	Single	9	1	11	1	1	5	160	Yes
4	Mobile Phone	3	Single	7	1	15	0	1	0	121	Yes
4	Mobile Phone	3	Single	6	1	14	0	1	3	120	Yes
4	Laptop & Accessory	5	Single	8	0	23	0	1	3	134	No

➤ Create a new column named 'ChurnStatus'. Set its value to "Churned" if the corresponding value in the 'Churn' column is 1, else assign "Active"

*When the Churn value=1;

```
5713 /* Create a new column named 'ChurnStatus'. Set its value to "Churned" if the corresponding value in the 'Churn' column is 1, else assign "Active"*/
5714 • ALTER TABLE customer_churn ADD COLUMN ChurnStatus VARCHAR(200);
5715 • UPDATE customer_churn SET ChurnStatus='Churned' WHERE Churn=1;
```

iceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress	Complain	OrderAmountHikeFromlastYear	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount	ComplaintReceived	ChurnStatus
	Laptop & Accessory	2	Single	9	1	11	1	1	5	160	Yes	Churned
	Mobile Phone	3	Single	7	1	15	0	1	0	121	Yes	Churned
	Mobile Phone	3	Single	6	1	14	0	1	3	120	Yes	Churned
	Laptop & Accessory	5	Single	8	0	23	0	1	3	134	No	Churned

*When the Churn value <> 1;

```
5713 /* Create a new column named 'ChurnStatus'. Set its value to "Churned" if the corresponding value in the 'Churn' column is 1, else assign "Active"*/
5714 • ALTER TABLE customer_churn ADD COLUMN ChurnStatus VARCHAR(200);
5715 • UPDATE customer_churn SET ChurnStatus='Churned' WHERE Churn=1;
5716 • UPDATE customer_churn SET ChurnStatus='Active' WHERE Churn<> 1;
```

iceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress	Complain	OrderAmountHikeFromlastYear	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount	ComplaintReceived	ChurnStatus
	Fashion	2	Single	3	0	15	0	1	3	207	No	Churned
	Mobile Phone	5	Single	8	0	22	1	1	4	142	No	Churned
	Fashion	4	Divorced	2	0	13	1	1	6	173	No	Active
	Laptop & Accessory	3	Divorced	3	1	20	0	4	5	123	Yes	Active

Column Dropping:

➤ Drop the columns "Churn" and "Complain" from the table

* Churn column removed from the table.

customer_churn

- Columns
- CustomerID
- Tenure
- PreferredLoginDevice
- CityTier
- WarehouseToHome
- PreferredPaymentMode
- Gender
- HoursSpentOnApp
- NumberOfDeviceRegistered
- PreferredOrderCat
- SatisfactionScore
- MaritalStatus
- NumberOfAddress
- Complain
- OrderAmountHikeFromlastY
- CouponUsed
- OrderCount
- DaySinceLastOrder
- CashbackAmount
- ComplaintReceived
- ChurnStatus

```
5713 /* Create a new column named 'ChurnStatus'. Set its value to "Churned" if the
5714 • ALTER TABLE customer_churn ADD COLUMN ChurnStatus VARCHAR(200);
5715 • UPDATE customer_churn SET ChurnStatus='Churned' WHERE Churn=1;
5716 • UPDATE customer_churn SET ChurnStatus='Active' WHERE Churn<> 1;
5717
5718 /*Drop the columns "Churn" and "Complain" from the table */
5719 • ALTER TABLE customer_churn DROP COLUMN Churn;
```

CustomerID	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender
50001	4	Mobile Phone	3	6	Debit Card	Female
50002	1	Mobile Phone	1	8	UPI	Male
50003	1	Mobile Phone	1	30	Debit Card	Male
50004	0	Mobile Phone	3	15	Debit Card	Male

* Complain column removed from the table.

```

5707
5708 /*Create a new column named 'ComplaintReceived' with values "Yes" if th
5709 • ALTER TABLE customer_churn ADD COLUMN ComplaintReceived VARCHAR(100);
5710 • UPDATE customer_churn SET ComplaintReceived = 'Yes' WHERE Complain = 1;
5711 • UPDATE customer_churn SET ComplaintReceived = 'No' WHERE Complain <> 1;
5712
5713 /* Create a new column named 'ChurnStatus'. Set its value to "Churned"
5714 • ALTER TABLE customer_churn ADD COLUMN ChurnStatus VARCHAR(200);
5715 • UPDATE customer_churn SET ChurnStatus='Churned' WHERE Churn=1;
5716 • UPDATE customer_churn SET ChurnStatus='Active' WHERE Churn<> 1;
5717
5718 /*Drop the columns "Churn" and "Complain" from the table */
5719 • ALTER TABLE customer_churn DROP COLUMN Churn;
5720 • ALTER TABLE customer_churn DROP COLUMN Complain;

```

Data Exploration and Analysis:

➤ Retrieve the count of churned and active customers from the dataset;

```

5722 /* Retrieve the count of churned and active customers from the dataset*/
5723 • SELECT Churnstatus, COUNT(*) AS Churn_and_Active FROM customer_churn GROUP BY Churnstatus;

```

Churnstatus	Churn_and_Active
Churned	948
Active	4680

➤ Display the average tenure and total cashbackamount of customers who churned.

*Average Tenure

```

5725 /*Display the average tenure and total cashback amount of customers who churned*/
5726 • SELECT AVG(Tenure) AS Tenure_average FROM customer_churn WHERE ChurnStatus='Churned';

```

Tenure_average
3.1762

*Total cashbackamount

```

5727 • SELECT SUM(CashbackAmount) AS Total_cashback FROM customer_churn WHERE Churnstatus= 'Churned';

```

Total_cashback
152030

➤ Identify the city tier with the highest number of churned customers whose preferred order category is Laptop & Accessory.

```
5729 /* Identify the city tier with the highest number of churned customers whose preferred order category is Laptop & Accessory.*/
5730 • SELECT CityTier, COUNT(*) AS churned_customers FROM customer_churn WHERE ChurnStatus = 'Churned' AND PreferredOrderCat = 'Laptop & Accessory' GROUP BY CityTier LIMIT 1;
```

CityTier	churned_customers
3	150

➤ Identify the most preferred payment mode among active customers;

The most preferred method is Debit card.

```
5732 /* Identify the most preferred payment mode among active customers*/
5733 • SELECT PreferredPaymentMode ,COUNT(*) AS Payment_mode FROM customer_churn WHERE ChurnStatus='Active' GROUP BY PreferredPaymentMode ;
```

PreferredPaymentMode	Payment_mode
E wallet	474
Debit Card	1956
Cash on Delivery	386
Credit Card	1522
UPI	342

➤ Calculate the total order amount hike from last year for customers who are single and prefer mobile phones for ordering.

```
5735 /*Calculate the total order amount hike from last year for customers who are single and prefer mobile phones for ordering.*/
5736 • SELECT SUM(OrderAmountHikeFromLastYear) AS Total_Order_Amount_Hike FROM customer_churn WHERE MaritalStatus = 'Single' AND PreferredOrderCat = 'Mobile Phone';
5737
```

Total_Order_Amount_Hike
12177

➤ Find the average number of devices registered among customers who used UPI as their preferred payment mode.

```
5738 /* Find the average number of devices registered among customers who used UPI as their preferred payment mode.*/
5739 • SELECT AVG(NumberOfDeviceRegistered) FROM customer_churn WHERE PreferredPaymentMode='UPI';
```

AVG(NumberOfDeviceRegistered)
3.7174

➤ Determine the city tier with the highest number of customers

```
5743 /* Determine the city tier with the highest number of customers*/
5744 • SELECT CityTier ,COUNT(*) AS customer_count FROM customer_churn GROUP BY CityTier ;
5745
5746
```

CityTier	customer_count
3	1720
1	3666
2	242

➤ Identify the gender that utilized the highest number of coupons.

```
5746 /* Identify the gender that utilized the highest number of coupons*/
5747 • SELECT Gender, SUM(CouponUsed) AS total_coupons FROM customer_churn GROUP BY Gender ORDER BY total_coupons DESC LIMIT 1;
```

Gender	total_coupons
Male	5629

➤ List the number of customers and the maximum hours spent on the app in each preferred order category.

```
5749 /*List the number of customers and the maximum hours spent on the app in each preferred order category*/
5750 • SELECT PreferredOrderCat,COUNT(*) AS customercount,MAX(HoursSpentOnApp) AS maxhoursspent FROM customer_churn GROUP BY PreferredOrderCat;
```

PreferredOrderCat	customercount	maxhoursspent
Laptop & Accessory	2050	5
Mobile Phone	2078	5
Others	264	4
Fashion	826	5
Grocery	410	4

➤ Calculate the total order count for customers who prefer using credit cards and have the maximum satisfaction score

```
5752 /*Calculate the total order count for customers who prefer using credit cards and have the maximum satisfaction score*/
5753 • SELECT SUM(OrderCount) AS total_tount FROM customer_churn WHERE PreferredPaymentMode = 'Credit Card' AND SatisfactionScore = (
5754     SELECT MAX(SatisfactionScore) FROM customer_churn );
```

total_tount
1122

➤ What is the average satisfaction score of customers who have complained

```
5756 /* What is the average satisfaction score of customers who have complained*/
5757 • SELECT AVG(SatisfactionScore)AS avg_satisfaction_score FROM customer_churn WHERE ComplaintReceived='Yes';
```

avg_satisfaction_score
2.9988

➤ List the preferred order category among customers who used more than 5 coupons.

```
5759 /* List the preferred order category among customers who used more than 5 coupons.*/
5760 • SELECT PreferredOrderCat ,COUNT(*) AS customer_count FROM customer_churn WHERE CouponUsed > 5 GROUP BY PreferredOrderCat;
5761
```

PreferredOrderCat	customer_count
Others	28
Laptop & Accessory	99
Fashion	89
Mobile Phone	45
Grocery	42

- List the top 3 preferred order categories with the highest average cashback amount.

```
5762 /* List the top 3 preferred order categories with the highest average cashback amount.*/
5763 • SELECT PreferredOrderCat,AVG(CashbackAmount) AS avg_cashback FROM customer_churn GROUP BY PreferredOrderCat ORDER BY avg_cashback DESC LIMIT 3;
```

PreferredOrderCat	avg_cashback
Others	304.4545
Grocery	266.2366
Fashion	210.4031

- Find the preferred payment modes of customers whose average tenure is 10 months and have placed more than 500 orders.

```
5765 /* Find the preferred payment modes of customers whose average tenure is 10 months and have placed more than 500 orders.*/
5766 • SELECT PreferredPaymentMode FROM customer_churn WHERE OrderCount > 500 AND Tenure = (SELECT AVG(Tenure) FROM customer_churn);
```

PreferredPaymentMode

- Categorize customers based on their distance from the warehouse to home such as 'Very Close Distance' for distances <=5km, 'Close Distance' for <=10km, 'Moderate Distance' for <=15km, and 'Far Distance' for >15km. Then, display the churn status breakdown for each distance category.

```
5770 • SELECT WarehouseToHome,ChurnStatus,
5771 CASE
5772 WHEN WarehouseToHome <= 5 THEN 'Very Close Distance'
5773 WHEN WarehouseToHome <= 10 THEN 'Close Distance'
5774 WHEN WarehouseToHome <= 15 THEN 'Moderate Distance'
5775 ELSE 'Far Distance'
5776 END AS Distance_Category FROM customer_churn;
5777
```

	WarehouseToHome	ChurnStatus	Distance_Category
	12	Churned	Moderate Distance
	22	Churned	Far Distance
	11	Churned	Moderate Distance
	6	Churned	Close Distance
	9	Churned	Close Distance
	31	Churned	Far Distance
	18	Churned	Far Distance

- List the customer's order details who are married, live in City Tier-1, and their order counts are more than the average number of orders placed by all customers.

```

5778 • SELECT CityTier,MaritalStatus,OrderCount FROM customer_churn WHERE MaritalStatus = 'Married' AND CityTier = 1 AND OrderCount > (SELECT AVG(OrderCount)FROM customer_churn);
5779

```

CityTier	MaritalStatus	OrderCount
1	Married	6
1	Married	4
1	Married	7
1	Married	3
1	Married	6
1	Married	3
1	Married	9

a) 'customer returns' table

	ReturnID	CustomerID	ReturnDate	RefundAmount
▶	1001	50022	2023-01-01	2130
	1002	50316	2023-01-23	2000
	1003	51099	2023-02-14	2290
	1004	52321	2023-03-08	2510
	1005	52928	2023-03-20	3000
	1006	53749	2023-04-17	1740
	1007	54206	2023-04-21	3250
	1008	54838	2023-04-30	1990

- b) Display the return details along with the customer details of those who have churned and have made complaints.

```

5793 • SELECT ch.ChurnStatus,ch.ComplaintReceived,cr.ReturnID,cr.CustomerID,cr.ReturnDate,cr.Refundamount
5794 FROM customer_returns cr
5795 LEFT JOIN customer_churn ch
5796 ON cr.CustomerID=ch.CustomerID
5797 WHERE Churnstatus='Churned' AND ComplaintReceived='Yes';
5798
5799

```

ChurnStatus	ComplaintReceived	ReturnID	CustomerID	ReturnDate	Refundamount
▶ Churned	Yes	1002	50316	2023-01-23	2000
Churned	Yes	1004	52321	2023-03-08	2510
Churned	Yes	1006	53749	2023-04-17	1740